PCA Detailed Theory Guide

Dimensionality Reduction with Principal Component Analysis

A clear, visual, step-by-step reference for Day 5 ML work

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Based on your uploaded educx Day 5 materials and tutor PCA notebook/script

How to use this guide

This PDF is designed as a complete clarity reference before notebook work. Read it once slowly, then use the tables and diagrams as a quick reminder while coding.

Contents at a glance

Source alignment from your uploaded materials

The educx MVD Day 5 page frames the topic as dimensionality reduction with PCA: high-dimensional datasets suffer from the curse of dimensionality, and PCA reduces dimensions while retaining maximum variance for visualisation and faster training. It also defines the Beginner, Intermediate and Expert exercises: Wine Quality PCA visualisation, RNA-Seq scree plot plus Logistic Regression, and PCA vs UMAP vs t-SNE with runtime and trustworthiness comparison.

The uploaded tutor PCA script explains PCA as transforming many variables into fewer new variables while keeping important information, using StandardScaler, creating PC1 and PC2 as new axes, checking explained variance, and using PCA to speed up Logistic Regression on image data. This guide uses those ideas but explains them in simpler, visual language.

1. Why PCA exists

Many real datasets have too many columns to understand directly. Wine Quality has several chemical measurements. RNA-Seq can have thousands of gene-expression features. When features become numerous, plots become impossible, training can become slow, and patterns can be hidden inside repeated or noisy information.

> Simple idea: PCA is a method for making a smaller, smarter version of a dataset while trying to keep the biggest patterns.

Important vocabulary

Term	Simple explanation	Example
Dimension	One feature or column.	alcohol, pH, density, gene_001
High-dimensional data	Data with many features.	RNA-Seq with thousands of gene columns
Dimensionality reduction	Reducing many features into fewer useful features.	11 wine features -> 2 PCA components
Curse of dimensionality	Too many features make data harder to visualise, slower to train, and sometimes noisier.	20,000 gene columns are difficult for simple plotting and modelling

Kid-like analogy

Imagine a school bag filled with 100 items. You cannot carry everything. PCA tries to pack the most important items into a smaller bag. You may lose some small items, but you keep the items that explain most of the bag.

Complete PCA workflow

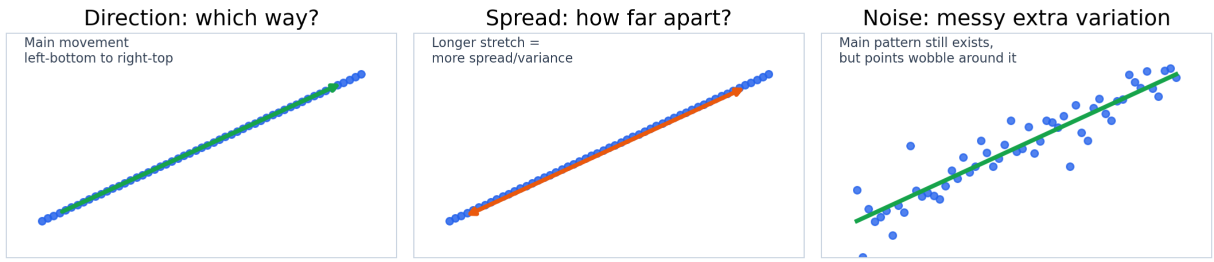
This is the practical PCA pipeline we will follow in the notebook.

The practical PCA workflow used in notebook projects.

2. Direction, spread, noise and variance

These words are the foundation of PCA. The sentence "PCA finds the direction with maximum variance" becomes easy only after these terms are clear.

Three foundation ideas before PCA



Direction, spread and noise shown as separate ideas.

Definitions

Concept	Simple explanation	Memory line
Direction	The way the data mainly moves.	Which way?
Spread	How far apart the values are.	How much stretch?
Noise	Messy, random variation around the main pattern.	Extra wobble around the pattern.
Variance	The mathematical measurement of spread.	Spread expressed as a number.

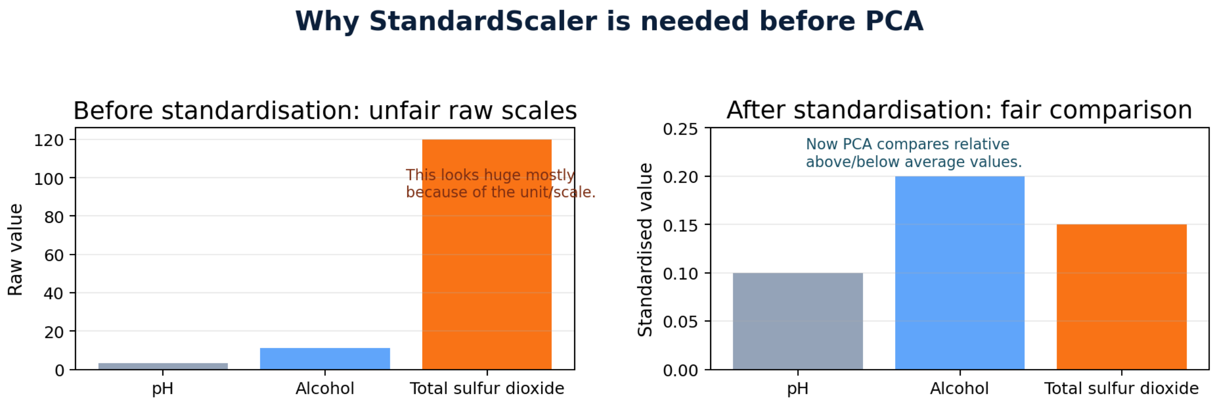
Example: study hours and exam score

If more study hours usually means a higher exam score, the main direction is upward. If scores are far apart, the spread is large. If some students perform higher or lower than expected because of sleep, stress or prior knowledge, that extra wobble is noise.

Core PCA sentence: PCA looks for the direction where the data is stretched the most. That direction becomes PC1.

3. Why standardisation is required before PCA

PCA is sensitive to scale. If one feature has much bigger raw numbers than another, PCA may think that feature is more important just because of the unit. This is dangerous.



Raw scales can fool PCA; standardisation gives features a fair comparison.

Wine example

Feature	Typical raw scale	Why this can be unfair
pH	Around 2.7 to 4.0	Small numbers do not mean unimportant.
Alcohol	Around 8 to 14	Moderate numbers.
Total sulfur dioxide	Can be much larger	May dominate PCA without scaling.
Chlorides	Often below 1	Small unit scale can be ignored unfairly.

What StandardScaler does

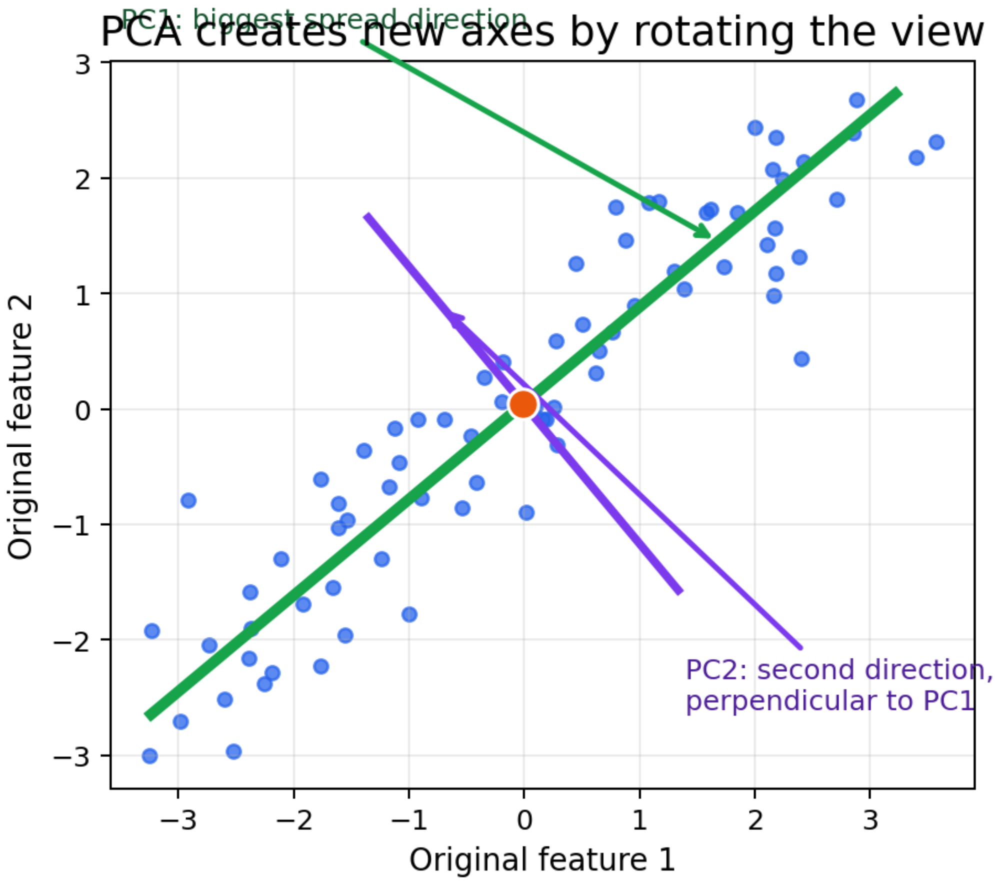
StandardScaler changes every feature so that its mean is 0 and its standard deviation is 1. In simple language, each value becomes "how far above or below normal it is for its own column".

```
# Practical PCA rule
X_scaled = StandardScaler().fit_transform(X)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

Never scale the target label. In Wine Quality, standardise the chemical input features X, not the quality label y.

4. Principal components: PC1, PC2 and linear combinations

A principal component is a new feature created by PCA. It is not an original column like alcohol or pH. It is a mixture of original features.



PCA rotates the axes: PC1 follows the biggest spread, PC2 is the next strongest direction.

PCs as recipes

Think of original features as ingredients. PCA creates new recipes. PC1 is the recipe that captures the most spread. PC2 is another recipe that captures the second-most spread.

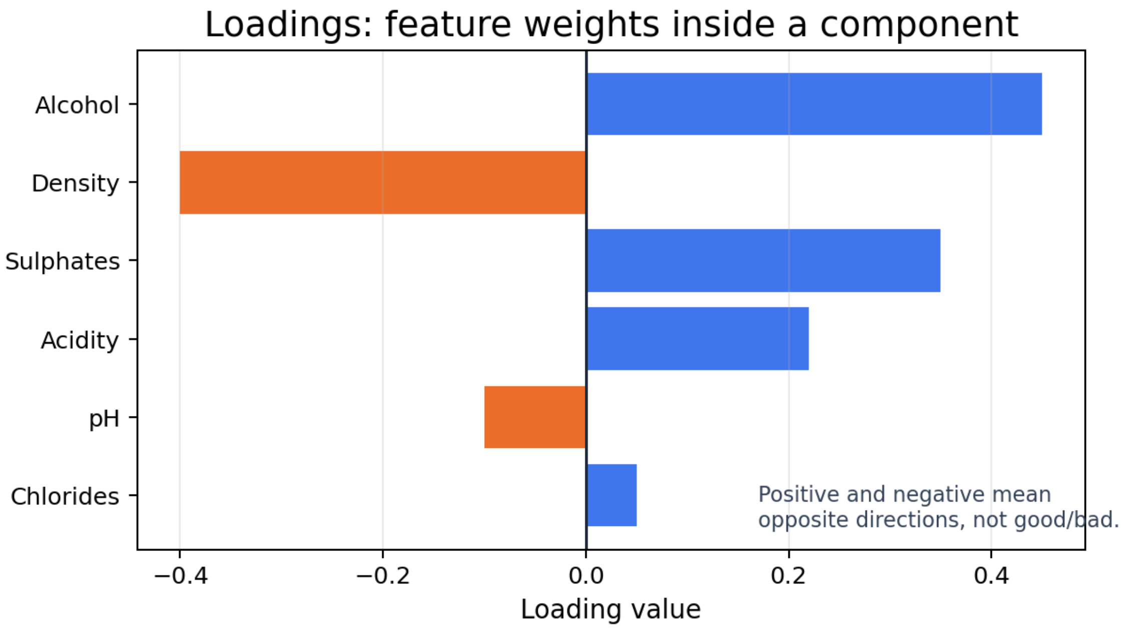
```
Example only:
PC1 = 0.45 * alcohol - 0.40 * density + 0.35 * sulphates + 0.20 * acidity
PC2 = -0.30 * alcohol + 0.50 * pH + 0.42 * chlorides - 0.25 * sulphates
```

Important properties

Property	Meaning
PC1	The component that captures the largest spread/variance.
PC2	The second-largest spread direction, perpendicular to PC1.
PC3 and later	Smaller remaining patterns.
Linear combination	A weighted mixture of original features.
Order	PCA ranks components by explained variance: PC1 first, then PC2, then PC3.

5. Loadings: how to interpret the components

Loadings are the weights inside a principal component. They tell us how strongly each original feature contributes to a component.



Example loadings: larger absolute values mean stronger contribution.

How to read loadings

Loading idea	Meaning
Large absolute value	The feature strongly contributes to that component.
Value close to 0	The feature contributes weakly.
Positive sign	Pushes the component in one direction.
Negative sign	Pushes the component in the opposite direction.
Positive vs negative	Not good vs bad. Only opposite directions.

Professional interpretation example

If PC1 has high positive loading for alcohol and high negative loading for density, a wine with a high PC1 score tends to have higher alcohol and lower density. This does not mean high PC1 is automatically good quality. PCA did not use the quality label.

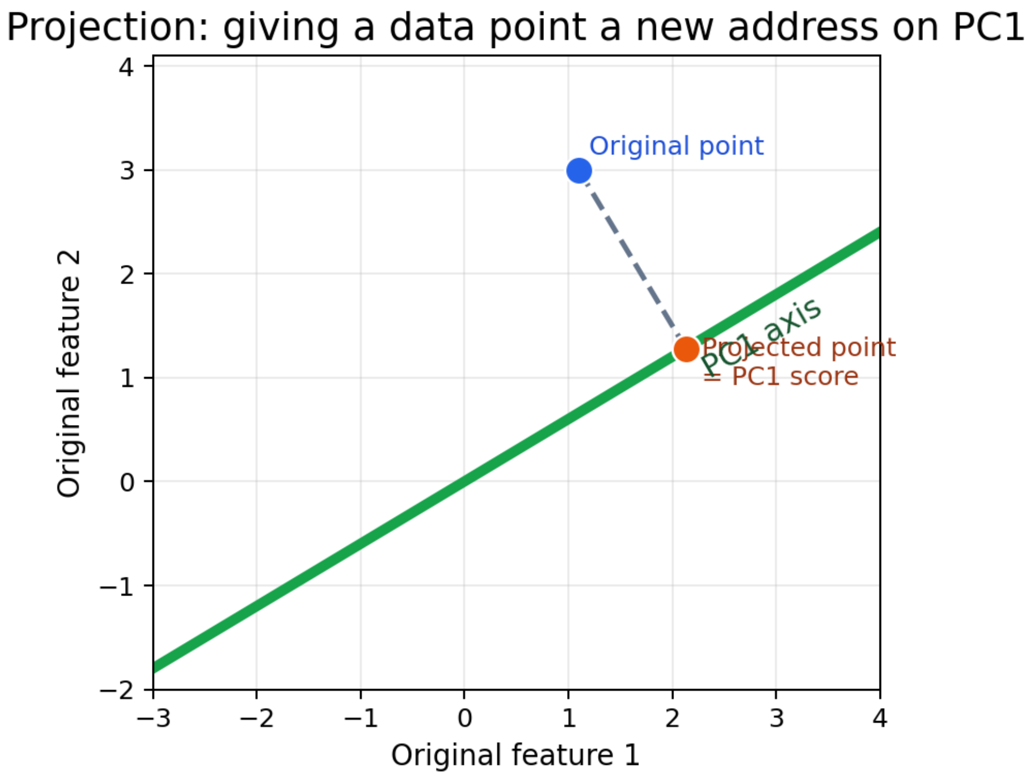
RNA-Seq connection

In the RNA-Seq task, loadings help identify which genes contribute most to PC1. Use absolute loading values to find the strongest contributors, then interpret positive and negative signs as opposite directions in gene-expression patterns.

> Memory line: Scores tell where rows land. Loadings tell how features build the axes.

6. Projection and PCA scores

Projection means placing each original data point onto the new PCA axes. After projection, each row gets new coordinates called PCA scores.



Projection: a data point is dropped onto the PC1 axis to get its PC1 score.

Original address vs PCA address

Before PCA	After PCA
Wine A = alcohol, pH, density, sulphates, chlorides, sugar, acidity, ...	Wine A = PC1 score, PC2 score
Patient A = thousands of gene-expression columns	Patient A = PC1 score, PC2 score, ...

Why projection matters

A PCA scatter plot is not drawn using the original features. It is drawn using the projected PCA scores. For a 2D PCA plot, the x-axis is PC1 score and the y-axis is PC2 score.

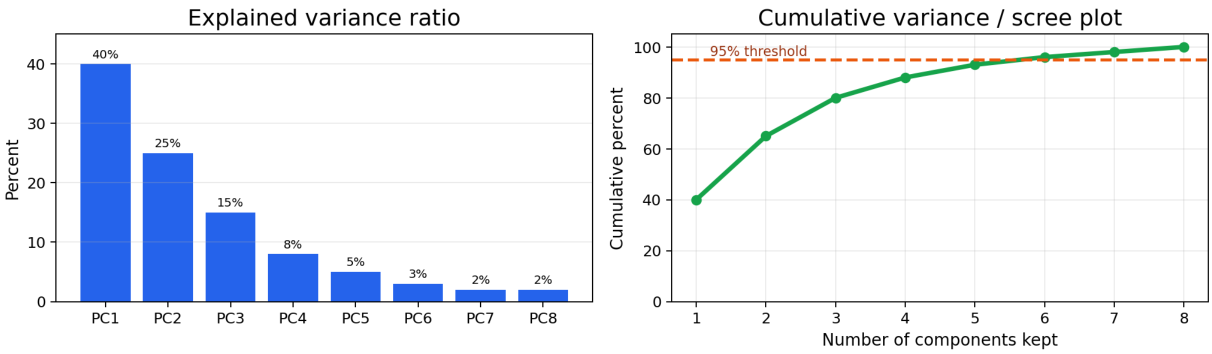
```
pc_df = pd.DataFrame({
    "PC1": X_pca[:, 0],
    "PC2": X_pca[:, 1],
    "quality": y
})
# Plot PC1 vs PC2 and colour by quality
```

Unsupervised reminder: PCA creates PC1 and PC2 without seeing the quality or tumour type label. Labels are used afterwards for colour.

7. Explained variance and scree plots

Explained variance tells us how much of the original data spread is captured by each PCA component. The explained variance ratio is the percentage share of total variance kept by each component.

How to know how much information PCA kept



Explained variance ratio and cumulative variance help decide how many components to keep.

Individual vs cumulative variance

Concept	Question it answers	Example
Explained variance ratio	How much does this one PC keep?	PC1 = 40%, PC2 = 25%
Cumulative explained variance	How much do PCs keep together?	PC1 + PC2 + PC3 = 80%
Scree plot	Where does adding more PCs stop helping much?	Curve rises fast, then flattens
95% rule	How many PCs keep at least 95%?	Choose the first component count reaching 95%

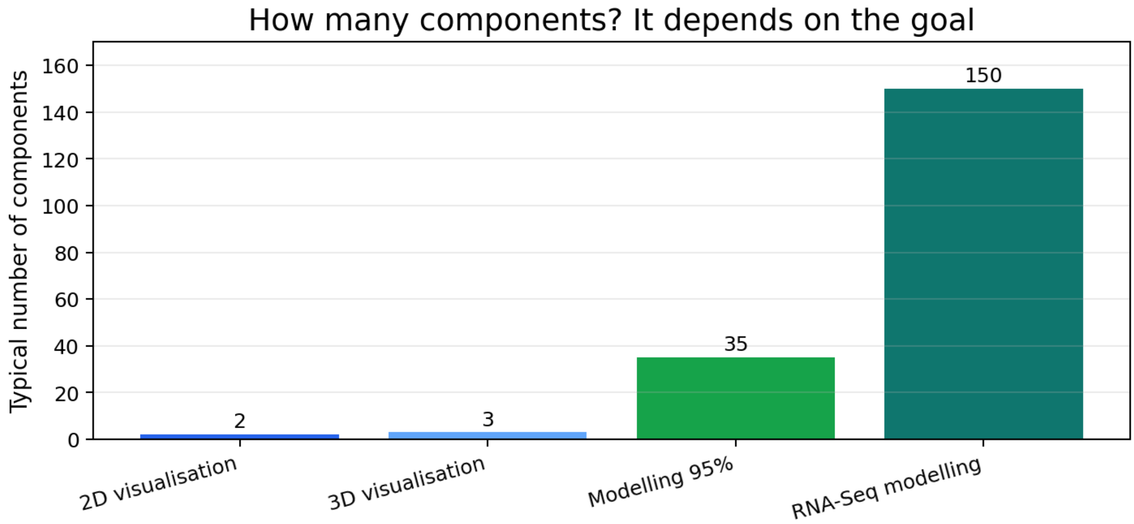
How to interpret common outputs

```
pca.explained_variance_ratio_
# Example output: array([0.32, 0.21])
# Interpretation: PC1 keeps 32%, PC2 keeps 21%, together 53%.
```

A 2D PCA plot can be useful even if PC1 + PC2 do not reach 95%. For visualisation, the goal is a readable picture. For modelling, we usually preserve more variance.

8. PCA for visualisation vs PCA for modelling

PCA has two major uses. First, it can create 2D or 3D plots from high-dimensional data. Second, it can reduce many columns before machine learning, often making training faster.



The number of components depends on whether the goal is visualisation or modelling.

Comparison

Goal	Typical components	Reason
2D visualisation	2	Humans understand x-y plots.
3D visualisation	3	Useful for richer plots, still human-readable.
Modelling compression	Enough for 90-95%+	Keep most information while reducing feature count.
RNA-Seq workflow	Calculated from scree plot	Thousands of genes need compression before modelling.

Model performance warning

PCA can speed up training and sometimes reduce noise, but it does not guarantee higher accuracy. PCA keeps variance, not necessarily the best target-predicting information. Therefore, the correct professional approach is to compare both accuracy and training time.

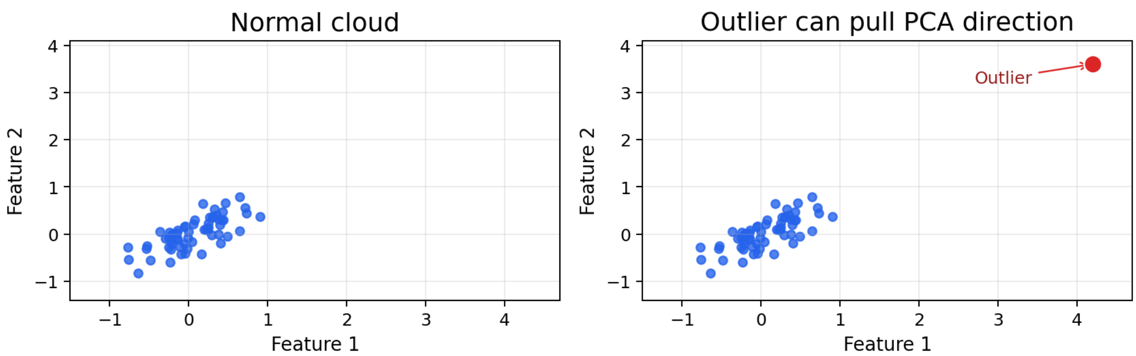
Connection to tutor material

The uploaded tutor PCA script demonstrates this idea with MNIST: scale data, choose PCA variance level such as 95%, transform train/test data, train Logistic Regression, then compare runtime and score across different retained variance levels.

9. PCA limitations and safe interpretation

PCA is powerful, but it is not magic. The most professional PCA reports are careful about what PCA can and cannot prove.

PCA follows spread, so outliers can matter a lot



Outliers can create artificial spread and distort PCA directions.

Limitations list

Limitation	Simple meaning	Safe interpretation rule
Linear method	PCA looks for straight directions.	Curved patterns may need UMAP/t-SNE.
Unsupervised	PCA does not use the target label.	Do not claim PCA optimised class separation.
Variance-focused	High variance may not equal prediction usefulness.	Always test model performance.
Information loss	Dropped PCs may contain useful detail.	Report explained variance.
Scale-sensitive	Raw large-number features can dominate.	Standardise first.
Outlier-sensitive	Extreme points create large spread.	Inspect outliers before PCA.
Harder to interpret	PCs are mixtures, not original features.	Use loadings for interpretation.

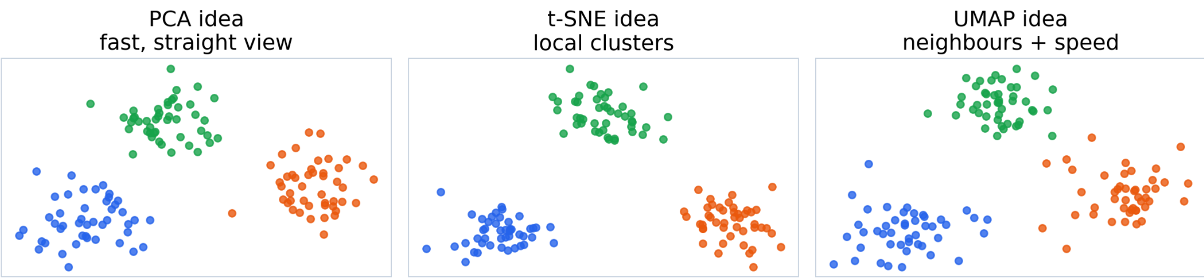
Professional wording template

> The PCA plot gives a simplified 2D view of the feature space. Some groups may show partial separation, but overlap is expected because PCA is unsupervised and preserves variance rather than directly optimising class separation.

10. PCA vs t-SNE vs UMAP

The Expert task compares PCA, t-SNE and UMAP on RNA-Seq data. All three can reduce high-dimensional data to 2D, but they think differently.

PCA vs t-SNE vs UMAP: different ways to make a 2D map



Illustrative comparison: PCA is fast and linear; t-SNE and UMAP focus more on local neighbourhoods.

Method comparison

Method	Simple idea	Strength	Caution
PCA	Best straight view.	Fast, interpretable variance, good baseline.	May miss curved/local structure.
t-SNE	Keep close neighbours close.	Often shows clear local clusters.	Can be slow; far distances can mislead.
UMAP	Keep neighbourhood structure efficiently.	Often good cluster structure and speed.	Still depends on parameters.
PCA + UMAP	Compress first, then use UMAP.	Can speed up and denoise high-dimensional data.	Must compare with direct UMAP.

Parameter intuition

Parameter	Method	Simple meaning
perplexity=30	t-SNE	Roughly controls how many neighbours t-SNE considers.
n_neighbors=15	UMAP	Controls local vs broader neighbourhood focus.
trustworthiness	Evaluation metric	Checks whether 2D neighbours were truly neighbours in original space.

Memory line: PCA = best straight view. t-SNE = local cluster view. UMAP = neighbour-preserving view, often faster than t-SNE.

11. Day 5 exercise map from uploaded MVD

The Day 5 document organises PCA learning into Beginner, Intermediate and Expert tasks. This page maps the theory to the exact practical work we will do in the notebook.

Level	Dataset	Task focus	Theory concepts used
Beginner	Wine Quality	Standardise data, PCA with 2 components, 2D scatter coloured by quality, explained variance bar chart, PCA with 3 components and 3D plot.	standardisation, projection, scores, explained variance, visualisation
Intermediate	Gene Expression RNA-Seq	PCA with all components, cumulative variance scree plot, find components for 95%, train LogisticRegression, compare accuracy and training time, heatmap of loadings.	scree plot, 95% rule, modelling PCA, runtime, loadings
Expert	Gene Expression RNA-Seq	Compare PCA, t-SNE and UMAP; measure runtime and trustworthiness; compare PCA(50) + UMAP with direct UMAP.	linear vs non-linear, local structure, trustworthiness, runtime trade-off

Notebook sequence we will follow

```
1. Add professional theory markdown
2. Import libraries and load Wine Quality
3. Inspect data and separate X/y
4. Standardise X
5. Apply PCA(n_components=2)
6. Plot PC1 vs PC2 coloured by quality
7. Plot explained_variance_ratio_
8. Apply PCA(n_components=3) and create 3D plot
9. Move to RNA-Seq: all-component PCA, scree plot, 95% threshold
10. Train LogisticRegression and compare accuracy/time
11. Interpret loadings and top genes
12. Expert comparison: PCA vs t-SNE vs UMAP
```

12. One-page quick reference

Use this page while coding.

Concept	Quick meaning
Dimension	A feature/column.
Direction	The way data mainly moves.
Spread / variance	How far apart the data is. Variance is the math measurement of spread.
Noise	Messy extra variation around the main pattern.
Standardisation	Make every feature fair: mean 0, standard deviation 1.
PC1	New PCA direction that captures the most variance.
PC2	Second strongest direction, perpendicular to PC1.
Linear combination	A weighted mixture of original features.
Loading	Feature weight inside a principal component.
Score	Row coordinate after PCA projection.
Projection	Placing original rows onto PCA axes.
Explained variance ratio	Percent of total spread kept by each PC.
Cumulative variance	Total spread kept after adding PCs together.
Scree plot	Plot used to decide how many PCs to keep.
95% rule	Keep the smallest number of components that reaches at least 95% variance.
PCA for visualisation	Use 2 or 3 components to draw a readable picture.
PCA for modelling	Use enough components to reduce features but preserve information.
PCA limitation	Linear, unsupervised, scale-sensitive, outlier-sensitive, can lose information.
t-SNE / UMAP	Non-linear methods that focus more on local neighbours/clusters.

13. Interpretation templates for your notebook

These are safe, professional sentence patterns you can adapt after running the notebook outputs.

For 2D PCA plot

> The 2D PCA plot shows the Wine Quality samples projected onto PC1 and PC2. The colours represent the quality labels, but PCA itself was fitted only on the input chemical features. Any separation should be interpreted as a simplified view of the feature space, not proof of perfect class separation.

For explained variance

> PC1 explains X% of the variance and PC2 explains Y%. Together, the 2D PCA plot preserves X+Y% of the original feature variance. This indicates how much of the full dataset structure is visible in the 2D representation.

For scree plot

> The cumulative explained variance curve reaches 95% at N components. Therefore, N components are sufficient to preserve at least 95% of the variance, reducing dimensionality while retaining most of the original structure.

For loadings

> The largest absolute loadings on PC1 indicate the original features/genes that contribute most strongly to the first principal component. Positive and negative signs indicate opposite directions along PC1, not good or bad effects.

For PCA vs UMAP vs t-SNE

> PCA provides a fast linear baseline, while t-SNE and UMAP focus more on preserving local neighbourhood structure. Runtime and trustworthiness help compare whether the visual improvements justify the computational cost.

14. Final simple story

PCA is like taking the best simplified photo of a complicated object.

If the object is complex, one photo cannot show everything. But a good angle can show the main shape. PCA finds that good angle mathematically by looking for the direction with the most spread. Then it creates new coordinates called principal components.

For Wine Quality, PCA helps us see many chemical features in 2D or 3D. For RNA-Seq, PCA helps reduce thousands of gene columns before modelling and comparison with non-linear methods.

The golden rules

Rule	Why it matters
Standardise before PCA.	Prevents big-number columns from dominating.
Use PC1/PC2 for visualisation, not full truth.	2D is a simplified map.
Always report explained variance.	Shows how much information was kept.
Use scree plot for modelling components.	Chooses a useful number of PCs.
Interpret loadings carefully.	They explain feature contribution, not prediction importance.
Compare accuracy and runtime.	PCA can help, but must be tested.
Remember PCA is unsupervised.	It does not know the target label.

You are now ready to start the notebook: Beginner Wine Quality PCA visualisation, one clean code cell at a time.